

MARIZFIRST

Backend Schema Document

Database Models, Prisma Schema, API Data Contracts, Indexes, Constraints, RBAC Rules, Audit Trails, Notifications and Operational Data Architecture

Item	Final Backend Direction
Product	MarizFirst - healthcare appointment discovery and assisted booking platform
Document type	Backend Schema Document / Database Architecture Specification
Source baseline	MarizFirst PRD v1.0 and TRD v1.0
MVP booking model	Hybrid booking: patient selects preferred slot; appointment starts as PENDING_CONFIRMATION until admin or partner action
Database	PostgreSQL with Prisma ORM and typed migrations
Primary backend concern	Reliable appointment lifecycle, partner-scoped access, auditability, SEO-ready public data and privacy-first patient data handling
Important rule	No discount/coupon data model is required for MVP. Fees can be stored, but no savings or coupon-led logic should drive the product.

Critical schema decision

The schema must treat MarizFirst as an operational appointment platform, not a static directory. The data model must support verified listings, slot selection, pending confirmation, admin/partner actions, WhatsApp notifications, source tracking, SEO metadata, RBAC ownership and audit logs from day one.

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1. Backend Schema Executive Summary

Backend mission

Design a clean relational backend that can power patient search, verified profiles, slot-based appointment requests, admin operations, partner dashboard, WhatsApp communication, SEO pages, analytics, auditability and secure role-based access without rebuilding the database after MVP.

Area	Schema Requirement	Reason
Discovery	Doctors, hospitals, clinics, specialties, cities, areas and blogs must be structured entities with SEO fields.	Public pages must be indexable and scalable city-by-city.
Booking	Appointment starts as PENDING_CONFIRMATION and references patient, provider target, optional slot and lead source.	Prevents fake auto-confirmation and supports assisted confirmation.
Operations	Admin and partner actions must be auditable and status-driven.	MarizFirst depends on operational trust and SLA handling.
Privacy	Patient phone, email, concern and notes must be restricted and never publicly exposed.	Healthcare appointment data is sensitive.
Partner access	Every partner-owned entity must be scoped by partner account or organization ownership.	Prevents partner data leakage.
Notifications	WhatsApp messages require templates, statuses, provider message IDs, retry count and webhook logs.	External notification failures must be visible and recoverable.

2. Schema Design Principles

Principle	Backend Meaning
Relational first	Use PostgreSQL relational integrity for appointment, partner, location and SEO data. Do not use a loose document-only database for the core operational system.
Status driven	Profiles, slots, appointments, leads, notifications and content should use explicit status fields. Avoid hidden states encoded only in notes.
Hybrid booking safe	Visible slots are selectable, but the system must not create a confirmed appointment until admin or partner action.
Ownership scoped	Partner dashboard data access must always be filtered by partner ownership at query level.
Audit everything sensitive	Role changes, exports, appointment status transitions, profile verification, notification retries and login failures must produce audit logs.
Soft delete by default	Doctors, hospitals, clinics, patients, appointments and partners should use status/deletedAt instead of destructive delete.
SEO ready	Slug, canonical, metadata, schema type and sitemap inclusion must be first-class fields on publishable entities.
No coupon logic in MVP	Store consultation fee where needed, but do not model discount campaigns or coupon promises as core product positioning.

3. Recommended Database Stack

Layer	Choice	Required Details
Database	PostgreSQL	Managed production database with automated backups, point-in-time recovery if available, and staging replica/isolated staging database.
ORM	Prisma	Typed schema, migrations, generated client, transaction support and clear relation mapping.
Migration flow	Prisma migrations	All schema changes committed through migration files. No manual production schema edits.
Validation	Zod + database constraints	Zod validates input; database constraints protect integrity. Never rely only on frontend validation.
IDs	CUID/UUID recommended	Use globally unique string IDs for easier distributed integration and safer public references.
Search	PostgreSQL indexes + optional full text later	MVP can use indexed name/slug/city/speciality queries; full-text or Meilisearch can be added after scale.
Queue	DB-backed queue or Redis/BullMQ	Notifications and reminders should not depend on request lifecycle.

4. Domain Model Overview

Domain	Entities	Purpose
Identity and RBAC	User, Role enum, PartnerProfile/Organization, Permission guard	Authenticate admins, content editors and partners; enforce ownership.
Healthcare listings	Doctor, Hospital, Clinic, Speciality, Facility, DoctorHospital, ClinicDoctor	Public searchable healthcare network.
Location	City, Area, Location	City-wise and area-wise discovery, local SEO and map/location details.
Booking	AppointmentSlot, Appointment, AppointmentStatusHistory	Preferred slot selection, pending confirmation and full lifecycle tracking.
Patient and lead	Patient, Lead, ConsentLog	Capture appointment request, source and consent safely.
Notifications	Notification, NotificationTemplate, WebhookEvent	WhatsApp acknowledgement, confirmation, reminders and failure recovery.
CMS and SEO	BlogPost, FAQ, SeoMetadata, Redirect, Sitemap inclusion fields	Content, dynamic metadata and programmatic SEO.
Commercial	ListingPlan, Commission, PartnerPackage	Listing fees, appointment commission and future monetization.
Security and operations	AuditLog, LoginEvent, ExportLog	Trace sensitive actions, failures, exports and operational accountability.

5. Core Enums

Enum	Values	Usage
UserRole	SUPER_ADMIN, ADMIN, CONTENT_EDITOR, DOCTOR_PARTNER, HOSPITAL_PARTNER, CLINIC_PARTNER	RBAC and dashboard routing.
UserStatus	ACTIVE, INVITED, SUSPENDED, DEACTIVATED	Login and account lifecycle.
PartnerType	DOCTOR, HOSPITAL, CLINIC, MULTI_ORG	Partner ownership and commercial account type.
PublishStatus	DRAFT, REVIEW, PUBLISHED, ARCHIVED	Public listing/content visibility.
VerificationStatus	UNVERIFIED, PENDING_REVIEW, VERIFIED, REJECTED, EXPIRED	Trust marker and profile approval.
AppointmentStatus	PENDING_CONFIRMATION, CONFIRMED, RESCHEDULED, CANCELLED, COMPLETED, NO_SHOW, LOST	Appointment lifecycle.
SlotStatus	AVAILABLE, HOLD, BOOKED, BLOCKED, EXPIRED	Slot selection and capacity logic.
LeadStatus	NEW, CONTACTED, CONFIRMED, LOST, COMPLETED, SPAM	Lead management and reporting.
NotificationChannel	WHATSAPP, EMAIL, SMS, SYSTEM	Communication channels.
NotificationStatus	QUEUED, SENT, DELIVERED, READ, FAILED, RETRYING, CANCELLED	Provider status tracking.
OwnerType	DOCTOR, HOSPITAL, CLINIC	Polymorphic slot/FAQ/media ownership.
AuditAction	CREATE, UPDATE, DELETE, RESTORE, STATUS_CHANGE, LOGIN_FAILED, EXPORT, VERIFY, PERMISSION_CHANGE	Audit log action classification.
PaymentStatus	PENDING, DUE, PAID, WAIVED, FAILED, CANCELLED	Commission/listing plan settlement tracking.

6. Entity Relationship Map

Relationship summary

A patient creates an appointment request against one target: doctor, hospital or clinic. The appointment can reference a selected slot. Appointment creation also creates a lead and queues notifications. Admin/partner actions update status history and audit logs.

Relationship	Cardinality	Backend Rule
User -> Doctor/Hospital/Clinic partner profile	1 to many or 1 to 1 depending partner type	Partner users can only manage assigned organization/listing records.
City -> Area -> Location	1 to many	Every public provider should resolve to city; area/location improves search and SEO.
Speciality -> Doctor	1 to many primary relation; optional many-to-many for multiple specialties later	MVP can support primarySpecialityId and optional DoctorSpeciality join.
Hospital -> Doctors	many to many	Use join table DoctorHospital with department and visibility fields.
Clinic -> Doctors/team	many to many	Use ClinicDoctor join table or simple clinic owner/team relation.
Doctor/Hospital/Clinic -> AppointmentSlot	1 to many via ownerType/ownerId	Slot owner must be validated before display/action.
Patient -> Appointment	1 to many	Patient matched by phone/email to avoid duplicates.
Appointment -> Lead	1 to 1 or 1 to many	One primary lead record for source tracking; additional lead activities can be separate.
Appointment -> Notifications	1 to many	Acknowledgement, partner alert, confirmation, reminder etc.
Appointment -> AppointmentStatusHistory	1 to many	Every status change stored with actor and reason.
Any sensitive entity -> AuditLog	1 to many polymorphic	Audit append-only and permission-protected.

7. User, Role and RBAC Schema

Table	Purpose	Critical Fields
users	Stores all human accounts: admins, content editors and partners.	id, name, email, phone, passwordHash/providerId, role, status, lastLoginAt, createdAt, updatedAt, deletedAt
partner_accounts	Commercial and ownership account for doctor/hospital/clinic partners.	id, userId, partnerType, organizationName, contactPerson, phone, email, status, onboardingStatus
partner_assignments	Allows one partner user to manage one or more healthcare listings.	id, userId, ownerType, ownerId, permissionLevel, active
sessions/accounts	Auth provider/session records if Auth.js/NextAuth is used.	provider, providerAccountId, userId, token fields
login_events	Tracks login success/failure for security audit.	id, userId, emailAttempted, status, ipAddress,

Table	Purpose	Critical Fields
		userAgent, reason, createdAt

Rule	Implementation Detail
Role check	Every admin/partner API endpoint must check authenticated user role before querying private data.
Ownership check	Partner query must include ownerType/ownerId or partnerUserId filters directly in WHERE clause.
Content editor restriction	Content editors can access content and SEO tables, not patient, appointment notes or finance data.
Export protection	Export requires explicit permission and creates ExportLog + AuditLog.
Soft-disable users	Suspend/deactivate accounts instead of deleting, because historical audit trails need actor identity.

8. Public Healthcare Listing Schema

Entity	Fields	Notes
Doctor	id, userId, name, slug, primarySpecialtyId, cityId, arealId, qualification, experienceYears, registrationNo, fee, bio, languages, gender, profileImageId, verificationStatus, publishStatus, seoTitle, seoDescription, canonicalUrl, schemaJson, createdAt, updatedAt, deletedAt	Public only when publishStatus=PUBLISHED and verificationStatus allows display. Registration number should be shown if available.
Hospital	id, partnerAccountId, name, slug, cityId, arealId, address, facilities, description, gallery, phone, mapUrl, verificationStatus, publishStatus, seo fields, createdAt, updatedAt, deletedAt	Can link multiple doctors and departments.
Clinic	id, partnerAccountId, name, slug, cityId, arealId, address, feeRange, facilities, description, gallery, phone, mapUrl, verificationStatus, publishStatus, seo fields, createdAt, updatedAt, deletedAt	Can link doctors/team and own slots.
Speciality	id, name, slug, description, icon, seoTitle, seoDescription, contentSections, publishStatus	Used for profile taxonomy, filters, SEO pages.
Facility	id, name, slug, icon, type, status	Reusable facility tags for hospital/clinic cards.
MediaAsset	id, url, provider, publicId, altText, width, height, mimeType, sizeBytes, ownerType, ownerId, createdAt	All public images need alt text and optimized delivery.
FAQ	id, ownerType, ownerId, question, answer, sortOrder, isPublished	Generates FAQ UI and FAQPage schema when relevant.

9. Location and Discovery Schema

Table	Fields	Backend Rules
cities	id, name, slug, state, country, seoTitle, seoDescription, introContent, publishStatus	City slug unique. Only published city pages appear in sitemap.
areas	id, cityId, name, slug, description, latitude, longitude, publishStatus	Area slug unique within city. Improves local filtering and landing pages.
locations	id, cityId, arealId, addressLine1, addressLine2, landmark, latitude, longitude, mapUrl, postalCode	Providers can reference normalized location or store simple address.
search_synonyms	id, keyword, targetType, targetId, cityId, active	Optional P1 feature for common misspellings and medical term aliases.

10. Slot and Appointment Schema

Entity	Required Fields	Rules
AppointmentSlot	id, ownerType, ownerId, startAt, endAt, timezone, capacity, bookedCount, status, createdById, createdAt, updatedAt, expiresAt	Only AVAILABLE with capacity > bookedCount can be selected. Past slots become EXPIRED.
Appointment	id, patientId, doctorId, hospitalId, clinicId, slotId, status, sourcePage, sourceType, concern, adminNotes, partnerNotes, requestedAt, confirmedAt, completedAt, cancelledAt, cancellationReason, createdAt, updatedAt	Exactly one target should be set: doctorId OR hospitalId OR clinicId. Initial status is PENDING_CONFIRMATION.
AppointmentStatusHistory	id, appointmentId, fromStatus, toStatus, actorId, actorRole, reason, note, createdAt	Required for every status change.
RescheduleRequest	id, appointmentId, oldSlotId, newSlotId, requestedById, reason, status, createdAt, resolvedAt	Optional if reschedule flow needs approval instead of immediate status change.

Appointment Status	Allowed Next Status	Required Side Effects
PENDING_CONFIRMATION	CONFIRMED, RESCHEDULED, CANCELLED, LOST	Create acknowledgement, admin alert, partner alert, SLA timer.

Appointment Status	Allowed Next Status	Required Side Effects
CONFIRMED	COMPLETED, CANCELLED, RESCHEDULED, NO_SHOW	Send confirmation message, update slot capacity/BOOKED if needed.
RESCHEDULED	CONFIRMED, CANCELLED, LOST	Store old/new slot, notify patient and partner.
CANCELLED	Closed	Reason recommended, notify relevant users, release slot if policy allows.
COMPLETED	Closed	Used for partner quality and commission.
NO_SHOW	Closed unless admin manually reopens	Reason/note recommended.
LOST	Closed	Reason required for reporting.

11. Patient and Lead Schema

Table	Purpose	Fields / Rules
patients	Stores patient identity captured through booking form.	id, name, phone, email, cityId, consentAt, consentSource, createdAt, updatedAt, deletedAt. Phone indexed.
consent_logs	Stores consent evidence for privacy and communication.	id, patientId, consentType, consentTextVersion, sourcePage, ipAddress, userAgent, consentAt.
leads	Tracks acquisition source and operational status.	id, patientId, appointmentId, sourcePage, sourceType, utmSource, utmMedium, utmCampaign, referrer, status, assignedAdminId, createdAt, updatedAt.
lead_activities	Tracks follow-ups beyond appointment status history.	id, leadId, actorId, activityType, note, createdAt.

Privacy rule

Do not send patient phone, email or health concern text to public analytics tools. Store concern only inside appointment data with restricted access.

12. Notification and WhatsApp Schema

Entity	Fields	Backend Rules
NotificationTemplate	id, name, channel, providerTemplateId, language, category, bodyPreview, variablesJson, status, createdAt	Templates require approval before production use. No medical advice or treatment guarantees.
Notification	id, appointmentId, patientId, partnerAccountId, recipientType, recipientPhone, channel, templateId, variablesJson, status, providerMessageId, retryCount, scheduledFor, sentAt, deliveredAt, readAt, errorMessage, createdAt	Notification creation should be queued. Provider failure must not silently disappear.
WebhookEvent	id, provider, eventType, providerMessageId, payloadJson, processedAt, processingStatus, errorMessage, createdAt	Webhook processing must verify secret/signature and be idempotent.
NotificationAttempt	id, notificationId, attemptNo, status, responseJson, errorMessage, attemptedAt	Useful for retry audit and support debugging.

13. Content, SEO and CMS Schema

Entity	Fields	Rules
BlogPost	id, title, slug, excerpt, content, categoryId, authorId, status, publishedAt, seoTitle, seoDescription, canonicalUrl, coverImageId	Published posts appear in sitemap and must avoid medical guarantee claims.
BlogCategory	id, name, slug, description, seo fields	Supports Health Guides taxonomy.
SeoMetadata	id, ownerType, ownerId, title, description, canonicalUrl, robots, schemaType, schemaJson, ogImageId, sitemapIncluded	Can be embedded on each entity or centralized. Use one strategy consistently.
Redirect	id, sourcePath, destinationPath, statusCode, active, createdAt	Required for slug changes and removed pages.
StaticPage	id, key, title, slug, content, status, seo fields	Privacy Policy, Terms, Medical Disclaimer, Partner Terms, Data Deletion.
HomepageSection	id, key, title, contentJson, sortOrder, status	Optional if homepage content must be editable by admin.

14. Revenue and Partner Commercial Schema

Entity	Fields	MVP Decision
ListingPlan	id, partnerAccountId, planType, monthlyFee, startDate, endDate, status, notes	Store commercial arrangement, but no payment gateway required in Phase 1.
Commission	id, appointmentId, partnerAccountId, amount, status, dueDate, paidAt, notes	Use after confirmed/completed appointment; do not automate payouts yet.
FeaturedPlacement	id, ownerType, ownerId, placementType, cityId, specialityId, startAt, endAt, status	Optional P1 for featured listings. Must not reduce trust or hide organic relevance.
PartnerInvoice	id, partnerAccountId, amount, status, periodStart, periodEnd, generatedAt, paidAt	P1/P2 if billing operations start.

15. Audit, Security and Compliance Schema

Table	Fields	Must Track
AuditLog	id, actorId, actorRole, action, entityType, entityId, beforeJson, afterJson, ipAddress, userAgent, requestId, createdAt	Create/update/delete/status changes, exports, role changes, verification changes, sensitive notes.
ExportLog	id, actorId, exportType, filtersJson, recordCount, fileId, status, createdAt	Any patient/lead/appointment export.
SecurityEvent	id, userId, eventType, severity, ipAddress, userAgent, metadataJson, createdAt	Suspicious login, rate limit triggers, permission denials.
DataDeletionRequest	id, patientId, requesterName, phone, email, status, requestedAt, resolvedAt, notes	Supports legal/privacy process.

16. Field-Level Database Tables

Implementation note

The following tables are written as an implementation blueprint. Final Prisma names may use PascalCase models, but the database meaning, relations, indexes and constraints should remain intact.

users

Field	Type	Constraint	Purpose
id	String / UUID	PK	Unique account ID
name	String	Required	Display name
email	String	Unique nullable	Admin/partner login email
phone	String	Unique nullable	Partner/admin phone if available
passwordHash	String	Nullable	If credentials auth is used
role	UserRole	Required	RBAC role
status	UserStatus	Default INVITED/ACTIVE	Account state
lastLoginAt	DateTime	Nullable	Security tracking
deletedAt	DateTime	Nullable	Soft delete

doctors

Field	Type	Constraint	Purpose
id	String	PK	Doctor listing ID
userId	String	FK nullable	Linked doctor partner user
name	String	Required	Doctor full name
slug	String	Unique	SEO URL
primarySpecialtyId	String	FK	Main speciality
cityId	String	FK	Main city
areaId	String	FK nullable	Area/locality
qualification	String	Required for publish	Degrees/credentials
experienceYears	Int	Nullable	Card/profile display
registrationNo	String	Nullable	Trust field
fee	Decimal/Int	Nullable	Consultation fee only; no discount logic
bio	Text	Nullable	Profile copy
languages	String[]/Json	Nullable	Spoken languages
gender	String enum	Nullable	Filter field
verificationStatus	VerificationStatus	Default UNVERIFIED	Trust marker
publishStatus	PublishStatus	Default DRAFT	Public visibility
seoTitle	String	Nullable	Metadata
seoDescription	String	Nullable	Metadata
deletedAt	DateTime	Nullable	Soft delete

hospitals

Field	Type	Constraint	Purpose
id	String	PK	Hospital ID
partnerAccountid	String	FK nullable	Owner/commercial partner
name	String	Required	Hospital name
slug	String	Unique	SEO URL
cityId	String	FK	City
areaId	String	FK nullable	Area
address	Text	Required for publish	Public address
description	Text	Nullable	Profile content
facilities	Json/String[]	Nullable	Facility tags
phone	String	Nullable	Public/operations number
mapUrl	String	Nullable	Google map link
verificationStatus	VerificationStatus	Required	Trust marker
publishStatus	PublishStatus	Required	Visibility
seoTitle	String	Nullable	Metadata
seoDescription	String	Nullable	Metadata

clinics

Field	Type	Constraint	Purpose
id	String	PK	Clinic ID
partnerAccountid	String	FK nullable	Owner/commercial partner
name	String	Required	Clinic name
slug	String	Unique	SEO URL
cityId	String	FK	City
areaId	String	FK nullable	Area
address	Text	Required for publish	Public address
feeRange	String	Nullable	Simple display range
description	Text	Nullable	Profile content
facilities	Json/String[]	Nullable	Facility tags
verificationStatus	VerificationStatus	Required	Trust marker
publishStatus	PublishStatus	Required	Visibility

appointment_slots

Field	Type	Constraint	Purpose
id	String	PK	Slot ID
ownerType	OwnerType	Required	DOCTOR/HOSPITAL/CLINIC
ownerId	String	Required	Listing ID
startAt	DateTime	Required indexed	Slot start time
endAt	DateTime	Required	Slot end time
timezone	String	Default Asia/Kolkata	Timezone
capacity	Int	Default 1	Max requests/bookings
bookedCount	Int	Default 0	Capacity tracking
status	SlotStatus	Default AVAILABLE	Selection state
createdById	String	FK nullable	Admin/partner creator
expiresAt	DateTime	Nullable	For HOLD strategy

appointments

Field	Type	Constraint	Purpose
id	String	PK	Appointment ID
patientId	String	FK	Patient
doctorId	String	FK nullable	Doctor target
hospitalId	String	FK nullable	Hospital target
clinicId	String	FK nullable	Clinic target
slotId	String	FK nullable	Preferred slot
status	AppointmentStatus	Default PENDING_CONFIRMATION	Lifecycle
sourcePage	String	Nullable	Page where submitted
sourceType	String	Nullable	doctor/hospital/clinic/city/blog/home
concern	Text	Nullable sensitive	Patient-provided concern
adminNotes	Text	Nullable private	Admin notes
partnerNotes	Text	Nullable scoped	Partner notes
requestedAt	DateTime	Required	Request created
confirmedAt	DateTime	Nullable	Confirmation time
completedAt	DateTime	Nullable	Completion time
cancelledAt	DateTime	Nullable	Cancellation time
cancellationReason	String	Nullable	Reason

patients

Field	Type	Constraint	Purpose
id	String	PK	Patient ID
name	String	Required	Patient name
phone	String	Required indexed	Primary identifier
email	String	Nullable indexed	Optional
cityId	String	FK nullable	Patient city
consentAt	DateTime	Required	Consent timestamp

Field	Type	Constraint	Purpose
createdAt	DateTime	Required	Created
updatedAt	DateTime	Required	Updated
deletedAt	DateTime	Nullable	Soft delete

notifications

Field	Type	Constraint	Purpose
id	String	PK	Notification ID
appointmentId	String	FK nullable	Related appointment
recipientType	String	Required	PATIENT/ADMIN/PARTNER
recipientPhone	String	Nullable	WhatsApp recipient
channel	NotificationChannel	Required	WHATSAPP etc.
templateId	String	FK nullable	Template
variablesJson	Json	Nullable	Template variables
status	NotificationStatus	Required	Queue/send status
providerMessageId	String	Indexed nullable	WhatsApp provider ID
retryCount	Int	Default 0	Retry
scheduledFor	DateTime	Nullable	Reminder schedule
sentAt	DateTime	Nullable	Sent
errorMessage	String	Nullable	Failure detail

audit_logs

Field	Type	Constraint	Purpose
id	String	PK	Audit ID
actorId	String	FK nullable	User/system
actorRole	String	Nullable	Role at time
action	AuditAction	Required	Action type
entityType	String	Required	Affected entity
entityId	String	Required	Affected ID
beforeJson	Json	Nullable	Previous state
afterJson	Json	Nullable	New state
ipAddress	String	Nullable	Security
userAgent	String	Nullable	Security
createdAt	DateTime	Required	Timestamp

17. Prisma Schema Blueprint

Prisma blueprint warning

This is a development-ready starting point, not the final generated migration. Developers should finalize naming, relation aliases and provider-specific settings during implementation.

Core Prisma schema starter

```
enum UserRole { SUPER_ADMIN ADMIN CONTENT_EDITOR DOCTOR_PARTNER HOSPITAL_PARTNER CLINIC_PARTNER }
enum PublishStatus { DRAFT REVIEW PUBLISHED ARCHIVED }
enum VerificationStatus { UNVERIFIED PENDING_REVIEW VERIFIED REJECTED EXPIRED }
enum AppointmentStatus { PENDING_CONFIRMATION CONFIRMED RESCHEDULED CANCELLED COMPLETED NO_SHOW LOST }
enum SlotStatus { AVAILABLE HOLD BOOKED BLOCKED EXPIRED }
enum OwnerType { DOCTOR HOSPITAL CLINIC }
enum NotificationChannel { WHATSAPP EMAIL SMS SYSTEM }
enum NotificationStatus { QUEUED SENT DELIVERED READ FAILED RETRYING CANCELLED }

model User {
  id          String    @id @default(cuid())
  name        String
  email       String?   @unique
  phone       String?   @unique
  passwordHash String?
  role        UserRole
  status      String    @default("ACTIVE")
  lastLoginAt DateTime?
  createdAt   DateTime  @default(now())
  updatedAt   DateTime  @updatedAt
  deletedAt   DateTime?

  auditLogs   AuditLog[]
}
```

```

model Doctor {
  id          String          @id @default(cuid())
  userId      String?
  name        String
  slug        String          @unique
  primarySpecialityId String
  cityId      String
  areaId      String?
  qualification String?
  experienceYears Int?
  registrationNo String?
  fee         Int?
  bio         String?
  languages   Json?
  gender      String?
  verificationStatus VerificationStatus @default(UNVERIFIED)
  publishStatus PublishStatus @default(DRAFT)
  seoTitle    String?
  seoDescription String?
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt
  deletedAt   DateTime?

  appointments Appointment[]

  @@index([cityId, publishStatus])
  @@index([primarySpecialityId, publishStatus])
  @@index([name])
}

model AppointmentSlot {
  id          String          @id @default(cuid())
  ownerType   OwnerType
  ownerId     String
  startAt     DateTime
  endAt       DateTime
  timezone    String          @default("Asia/Kolkata")
  capacity    Int             @default(1)
  bookedCount Int             @default(0)
  status      SlotStatus @default(AVAILABLE)
  createdById String?
  expiresAt   DateTime?
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt

  appointments Appointment[]

  @@index([ownerType, ownerId, startAt])
  @@index([status, startAt])
}

model Appointment {
  id          String          @id @default(cuid())
  patientId   String
  doctorId    String?
  hospitalId  String?
  clinicId    String?
  slotId      String?
  status      AppointmentStatus @default(PENDING_CONFIRMATION)
  sourcePage  String?
  sourceType  String?
  concern     String?
  adminNotes  String?
  partnerNotes String?
  requestedAt DateTime @default(now())
  confirmedAt DateTime?
  completedAt DateTime?
  cancelledAt DateTime?
  cancellationReason String?
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt

  patient Patient @relation(fields: [patientId], references: [id])
  doctor  Doctor?  @relation(fields: [doctorId], references: [id])
  slot    AppointmentSlot? @relation(fields: [slotId], references: [id])
}

```

```

    lead          Lead?
    notifications Notification[]
    statusHistory AppointmentStatusHistory[]

    @@index([status, createdAt])
    @@index([patientId])
    @@index([doctorId])
    @@index([hospitalId])
    @@index([clinicId])
  }

model Patient {
  id          String    @id @default(cuid())
  name        String
  phone       String
  email       String?
  cityId      String?
  consentAt   DateTime
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt
  deletedAt   DateTime?

  appointments Appointment[]

  @@index([phone])
  @@index([email])
}

```

18. Indexing Strategy

Table	Indexes	Why
doctors	unique(slug), index(cityId,publishStatus), index(primarySpecialtyId,publishStatus), index(name)	SEO pages, filters, search and public profile routing.
hospitals/clinics	unique(slug), index(cityId,publishStatus), index(arealId), index(name)	Location search and listing performance.
appointment_slots	index(ownerType,ownerId,startAt), index(status,startAt)	Fast slot lookup and expiry jobs.
appointments	index(status,createdAt), index(patientId), index(doctorId/hospitalId/clinicId), index(slotId)	Admin inbox, partner dashboard, patient history and reporting.
patients	index(phone), index(email)	Patient upsert and duplicate reduction.
leads	index(status,createdAt), index(sourcePage), index(utmSource,utmMedium,utmCampaign)	Funnel reporting and lead operations.
notifications	index(status,scheduledFor), index(providerMessageId), index(appointmentId)	Queue processing, webhook matching and dashboard failure view.
audit_logs	index(actorId,createdAt), index(entityType,entityId), index(action,createdAt)	Security review and change history.
blog/city/speciality	unique(slug), index(publishStatus)	Public routing and sitemap generation.

19. Constraints and Validation Rules

Rule	Database / Service Enforcement
Appointment must have exactly one target	Service validation: only one of doctorId, hospitalId, clinicId may be non-null. Optional DB check constraint can be added manually.
Slot cannot overbook	Transactional check capacity > bookedCount before creating appointment/hold. Use row-level lock strategy if high concurrency appears.
Public profile cannot publish without basics	Admin publish validation: name, slug, city, speciality/facilities/address, verification state and SEO basics required.
Slug uniqueness	Unique database constraints for doctor/hospital/clinic/city/speciality/blog slugs.
Status transitions	Service layer state machine. Route handlers cannot directly set arbitrary status.
Partner ownership	Permission guard + scoped WHERE queries. Never fetch all then filter in UI.
Sensitive notes	adminNotes visible to admins only; partnerNotes visible to scoped partner/admin; concern restricted to appointment operations.
Webhook idempotency	providerMessageId or event ID unique handling to prevent duplicate status updates.

20. Soft Delete and Data Lifecycle

Data Type	Delete Strategy	Reason
Doctors/hospitals/clinics	deletedAt + publishStatus ARCHIVED	Preserve appointment history and SEO redirect options.
Patients	deletedAt/anonymization based on data deletion request policy	Privacy compliance while preserving non-identifying operational history where lawful.
Appointments	Never hard-delete in normal admin UI	Required for audit, partner billing and trust operations.
Audit logs	Append-only; no normal deletion	Security accountability.
Notifications	Retain provider status and failure logs for operational debugging	Support traceability.
Media	Do not immediately delete if used by published content	Avoid broken public pages.

21. API Data Contracts

Endpoint	Request Data	Response / Side Effects
POST /api/appointments	profileType, profileId, slotId, patientName, phone, cityId, optional email, optional concern, consent=true, sourcePage, UTM	Creates/updates Patient, creates Appointment PENDING_CONFIRMATION, creates Lead, queues notifications, writes audit log.
GET /api/search	q, city, speciality, type, page, limit	Returns typed results: doctors/hospitals/clinics/specialities/cities with safe public fields only.
POST /api/admin/appointments/:id/confirm	optional finalSlotId, note	Validates transition, confirms appointment, queues patient notification, writes status history + audit log.
POST /api/admin/appointments/:id/reschedule	newSlotId, reason, note	Validates new slot, updates status/history, notifies patient/partner.
POST /api/partner/slots	ownerType, ownerId, startAt, endAt, capacity, repeat/bulk options	Creates scoped slots after permission and conflict validation.
POST /api/webhooks/whatsapp	provider webhook payload	Verifies signature, stores WebhookEvent, updates Notification status idempotently.

22. Seed Data Requirements

Seed Category	Minimum Data	Notes
Roles/admin	1 super admin, 1 admin, 1 content editor	Use secure credentials/env-managed setup.
Cities/areas	Launch city + key areas	Example: Gurgaon and main healthcare areas if launch market is Gurgaon.
Specialities	10-20 launch specialities	Cardiology, Dermatology, Pediatrics, Gynecology, Orthopedics, ENT, etc.
Doctors	Verified real profiles only for production	No fake doctors. Staging can use clearly marked demo data.
Hospitals/clinics	Verified launch partners	Include address, city, facilities, photos, SEO metadata.
FAQ/legal	Privacy, Terms, Medical Disclaimer, Partner Terms, Data Deletion	Mandatory before production launch.
Notification templates	Acknowledgement, request alert, confirmation, reschedule, cancellation, reminder	WhatsApp templates require provider approval.

23. Migration Strategy

Stage	Action	Gate
Local	Develop schema changes with Prisma migration and seed data.	Migration runs cleanly from empty DB.
Review	Check generated SQL for destructive changes, index names and relation constraints.	No accidental column drops or data loss.
Staging	Apply migration to staging, run seed/test data and E2E booking flow.	All P0 workflows pass.
Production	Back up database, apply migration during low-traffic window, verify app health.	Rollback plan ready.
Post-migration	Run smoke tests: search, profile, booking, admin confirm, partner view, notifications.	No critical errors in monitoring.

24. Backup, Restore and Data Retention

Requirement	Implementation
Automated backups	Daily managed PostgreSQL backup at minimum; point-in-time recovery preferred.
Restore testing	Restore staging/test database periodically to verify backups are usable.
Private data handling	Backups contain sensitive patient data; access must be restricted.
Retention policy	Define retention for patients, leads, notifications, audit logs and webhook payloads before scale.
Export files	Temporary export files should expire and be access-controlled.

25. Security, Privacy and Access Rules

Risk	Schema/Backend Control
Broken partner access	Partner assignments and query-level ownership filters. Automated tests for URL/API manipulation.
Patient data exposure	Separate public profile fields from appointment/patient tables. No public API returns phone/email/concern.
Admin misuse	AuditLog and ExportLog for sensitive operations. Least privilege roles.
Spam leads	Rate-limit booking endpoint; optional OTP/CAPTCHA fields can be added without schema rebuild.
Secrets leakage	No provider tokens in DB unless encrypted. Use environment secret manager.
Webhook spoofing	WebhookEvent includes provider verification; reject unsigned/untrusted payloads.

26. Analytics and Event Storage

Event	Store Internally?	Fields
search_performed	Optional internal + GA4	query, city, speciality, result_count, no patient identifiers.
profile_viewed	Optional internal + GA4	profileType, profileId, city, speciality, sourcePage.
slot_selected	Optional internal	slotId, profileType, profileId, date/time.
appointment_submitted	Yes internal + safe GA4	appointmentId, profileId, sourcePage, status. Do not send phone/email/concern to GA4.
appointment_confirmed	Yes internal	appointmentId, partnerId, timeToConfirm.
appointment_cancelled/rescheduled	Yes internal	appointmentId, reasonCategory, actorRole. Avoid sensitive notes.

27. QA Test Data and Acceptance Scenarios

Scenario	Expected Backend Result
Patient submits booking with valid slot	Patient upserted, Appointment created PENDING_CONFIRMATION, Lead created, notifications queued, audit log written.
Patient submits invalid phone/no consent	Request rejected; no appointment created.
Two users select last-capacity slot concurrently	Only allowed capacity succeeds; no overbooking.
Partner tries to open another partner appointment	403/404 scoped denial; security event/audit as appropriate.
Admin confirms pending appointment	Status history created, confirmedAt set, patient confirmation notification queued.
WhatsApp webhook arrives twice	Webhook stored/recognized idempotently; notification status not duplicated incorrectly.
Content editor attempts patient export	Denied by RBAC; audit/security event recorded.
Doctor slug changes	Redirect created from old URL to new URL.

28. Open Decisions

Decision	Why It Matters	Recommended Default
Payload CMS vs custom admin	Affects schema ownership, workflow customization and speed.	Custom admin if appointment operations are complex; Payload if content/listing CRUD speed is priority.
Phone OTP at launch	Reduces spam but adds friction.	Start with rate limit + CAPTCHA; add OTP if spam appears.
Slot HOLD strategy	Determines whether pending requests reserve capacity.	Use HOLD with short expiry only if capacity is limited and partners respond fast; otherwise allow

Decision	Why It Matters	Recommended Default
Multi-speciality doctors	Some doctors may have multiple specialties.	capacity-based pending requests. Start with <code>primarySpecialtyId</code> + optional join table if needed.
Partner edit approvals	Controls data quality and trust.	Require admin review for sensitive fields: fee, registration, name, location, verification, SEO-critical info.
Data retention periods	Compliance and storage policy.	Define before launch; keep audit logs long-term.

29. Backend Schema Definition of Done

- All P0 entities, enums, relations, indexes and constraints are represented in Prisma schema and migrations.
- Appointment creation works transactionally and always starts as `PENDING_CONFIRMATION`.
- Partner ownership checks are enforced at backend query level for every partner dashboard/API route.
- Patient phone, email, concern and notes are never returned by public APIs.
- Admin can manage doctors, hospitals, clinics, slots, appointments, leads, notifications, content, SEO and audit history.
- Notification records store queue state, provider message ID, retries, failures and webhook updates.
- SEO fields exist for publishable public entities and sitemap can be generated from published records.
- Audit logs exist for sensitive operations and status changes.
- Soft delete strategy is implemented for business-critical records.
- Migrations run cleanly on local and staging databases with seed data.
- Indexes support launch search, listing, dashboard and notification queries.
- QA scenarios for overbooking, RBAC bypass, webhook idempotency and appointment lifecycle pass.